Advanced Database

Dr. M. Ali Nematbakhsh

Project2: Big Data

Section 1: Descriptive Questions
In this section, you must write complete descriptive answers based on your understanding

Question 1: Explain where the concept of Big Data originated & describe in detail the 3Vs of Big Data. Try to include examples.

Question 2: Based on the Hadoop file system architecture, please provide a comprehensive and detailed explanation of how it works. You should understand completely how NameNode, DataNode, and Replications work. Also, analyse the NameNodes as a single point of failure.

Question 3: HDFS performs poorly with a very large number of small files. Analyse the root causes.

Question 4: What is MapReduce, and what are the Advantages?
Explain different phases of the MapReduce Pipeline (Map, Shuffle & Sort, Reduce)
What are the limitations of MapReduce?
It would be great if you provide an easy example for the answer

Question 5: In the attached file, there is a Python program called "Simple_MapReduce.py". In this file, I wrote a simple MapReduce code. By making changes to the code, specify the output of each stage of the MapReduce and explain how it works.

Question 6: Suppose you need to store a very large number of small files, each of size, say, 2 kilobytes. If your choice is between a distributed file system and a distributed key-value store, which would you prefer, and explain why.

Section 2: Practical Questions
In this section, you'll face hands-on questions (I highly recommend using Jupyter Notebook for this section)

Question 1: You have joined the data engineering team at an online retail company. Recently, the company's CEO noticed that total sales don't add up correctly across countries, and he wants a detailed summary generated directly from the raw transaction data.
As the Head of this project, you want to calculate the total sales for each country.
Write a complete MapReduce Program with all phases (Map, Sort & Shuffle, Reduce) and provide us with the output.
The Link to the Dataset (E-Commerce Dataset) is provided on the next page.
Total sales must be calculated from the related columns (Please look at the Dataset and find those columns 😉)

Question 2: Generate a synthetic dataset with 50 records (Use ChatGPT, Grok, etc) that simulates Twitter posts. Each record must include the following fields :

- hashtags: A list of hashtags used in the post
- likeCount: Number of likes for the post
- retweetCount: Number of retweets
- replyCount: Number of replies

Use the MapReduce workflow to find the 3 Top trending hashtags. **The popularity of a post is identified as the sum of its likes, replies, and retweets 🙄**

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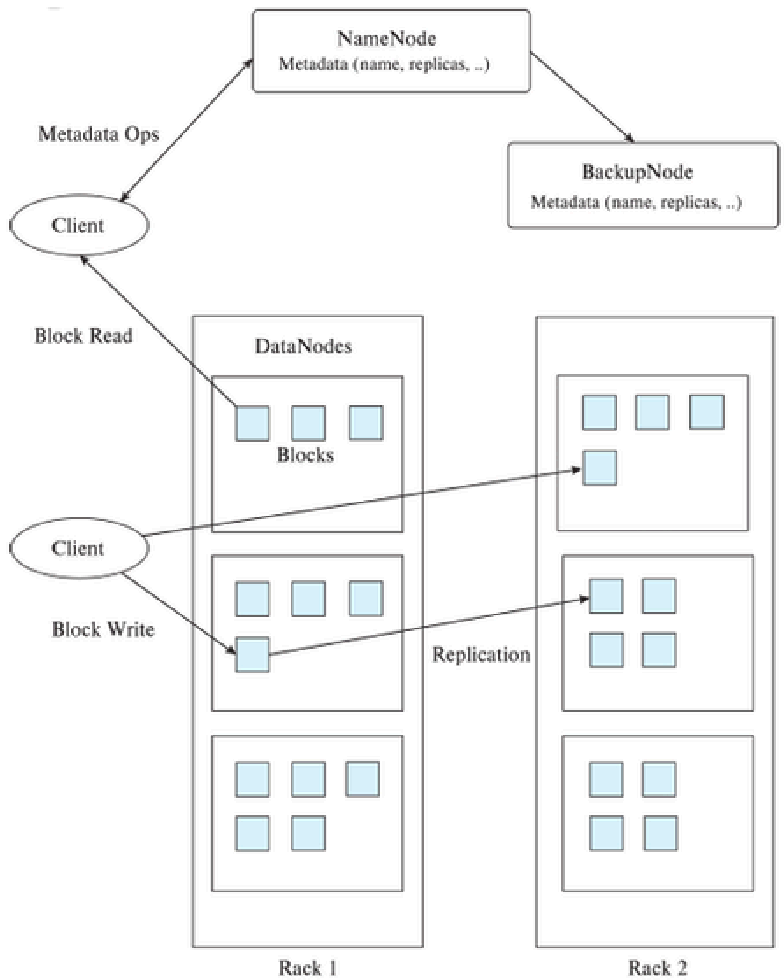
For a better understanding of the concepts and to answer the questions, refer to the links and information provided.

1: Hadoop File System Architecture :



2 : E-Commerce Dataset Link :
https://www.kaggle.com/datasets/carrie1/ecommerce-data

3: To understand the concepts better, I recommend to take a look at these sources:

- **https://www.kaggle.com/discussions/general/22338**
- **https://www.kaggle.com/discussions/getting-started/97824**
- **https://stackoverflow.com/questions/12375761/good-mapreduce-examples**
- **https://hadoop.apache.org/docs/stable/hadoop-mapreduce-client/hadoop-mapreduce-client-core/MapReduceTutorial.html**
- **https://datascienceguide.github.io/map-reduce**

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What to upload for us?

- Complete Documentation about answering the section 1 questions
- Complete Notebooks or program files from section 2 + Documentations
- Improved Simple_MapReduce.py file, related to Question 5 of section 1, with full documentation
- Generated Dataset by AI
- DO NOT UPLOAD E-COMMERCE DATASET

all your files must be in one ZIP file named:
DBHW{Project Number}-{Student Complete name}-{Student Number}

Students must follow the Template structure, or they will receive a grade deduction.

Dead Line : 29 November - 8 Azar

Behnam Soufi
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